

Definition of a new morphological parameter to improve prediction of urban heat island

Victor Equere, Parham A. Mirzaei*, Saffa Riffat

Architecture and Built Environment Department, University of Nottingham, Nottingham, UK

ARTICLE INFO

Keywords:

Urban heat island
Predictive tool
Remote-sensing
Regression
GIS
Morphology

ABSTRACT

The impact of the urban heat island (UHI) on health of urban residents, energy consumption, and urban air quality has become increasingly significant in cities. Despite of multiple recent studies on developing UHI predictive models, the impact of vertical morphological parameters on the UHI formation is barely investigated. The current work investigates the extent of vertical morphological components impact on the UHI intensity and proposes a novel vertical urban morphological parameter defined as the elevation factor. Using remote-sensing satellite images and a GIS based approach, both horizontal and vertical morphological components of San-Francisco, as the case study city, was investigated for 20 representative built up areas. High-resolution geometrical data of these areas were then correlated with their land surface temperatures (LST). Linear regression analysis and ANOVA test revealed that the model with the newly proposed elevation factor has a significant impact on representing of the UHI behaviour with a coefficient of determination factor (R^2) of 0.68. The model was then successfully validated with the UHI prediction of 24 randomly selected points from each of the five new areas. Furthermore, the impact of elevation morphology on the UHI intensity was investigated as the elevation factor was seen to provide a stronger correlation between urban morphology and UHI spatial distribution.

1. Introduction

Urban Heat Island (UHI) has a significant impact on health of urban residents (Mavrogianni, Wilkinson, Davies, Biddulph, & Oikonomou, 2012; Mirzaei et al., 2012; Morabito et al., 2015), energy consumption (Polydoros & Cartalis, 2015; Santamouris, 2007), as well as air quality (Filleul et al., 2006; Liu & Weng, 2012). The UHI research has shown that the urban morphological composition such as increased impervious land cover, geometry of street canyons, reduced sky visibility are the main key players, impacting on the energy balance, wind speed, and long-wave radiation, and resulting in the elevation of temperature in cities (Lima Alves & Lopes, 2017; Li, Zhang, Mirzaei, Zhang, & Zhao, 2018; Mirzaei, 2015; Nakata-Osaki, Souza, & Rodrigues, 2018; Oke, 1982). Numerous studies have given attention to improve techniques in studying of the UHI (Mirzaei & Haghighat, 2010; Mirzaei, Olsthoorn, Torjan, & Haghighat, 2015), in a bid to not just achieve healthier urban environments, but also more energy efficient buildings in cities (Ashtiani, Mirzaei, & Haghighat, 2014; Priyadarsini, 2009). In view of this, the thermal remote-sensing method have been adopted to investigate the impact of various morphological features on the UHI intensity of a region as well as the associated potential mitigation strategies.

Ever since the first remote-sensing studies on the urban surface temperature such as the one conducted by (Rao, 1972), the use of satellite or other airborne remote sensing approaches in the study of urban heat island has received increasing patronage by scientist, seeking to explore the urban heat island distribution in a large scale as this is necessary in acquiring morphological information for large spatial extents (Gaitani, Burud, Thiis, & Santamouris, 2017; Li, Zhou et al., 2018; Meng et al., 2018; Streutker, 2003). In many related studies, spatial resolution images obtained from satellite or airborne remote-sensing is used to assess the thermal behaviour of urban surfaces by correlating several indices and surface characteristics to the land surface temperature (LST) as one of the main indicators for UHI. A number of indices has since been developed to extract the horizontal morphological features from a Landsat image, which are explained in the below paragraph. It should be noted that despite of many developed indices and their advantageous in understanding horizontal spatial morphological characteristics of a city, these indices are barely developed to scrutinize the vertical characteristics of a city area.

Urban main built up areas (UMBAs) and impervious surface distribution density (ISDD) were mainly used to better differentiate sub-urban boundaries and to ensure the integrity of land cover types (Meng et al., 2018). (He, Shi, Xie, & Zhao, 2010) also proposed built-up index

* Corresponding author at: University Park, Nottingham, NG7 2RD, UK.
E-mail address: Parham.Mirzaei_Ahramjani@nottingham.ac.uk (P.A. Mirzaei).

Nomenclature

Abbreviations

ASTER	Advanced space borne thermal emission and reflection radiometer
BU	Built-up index
CCD	Charge-coupled device
DN	Digital number
DTM	Digital terrain model
EF	Elevation factor
GIS	Geographical information system
GPS	Global positioning satellite system
GVF	Ground view factor
IMU	Inertia navigational measurement unit
LANDSAT	Land remote-sensing satellite (system)
LiDAR	Light detection and ranging
LST	Land surface temperature
MLR	Multi-linear regression
NDBI	Normalized difference built-up index
NDVI	Normalized difference vegetation index
NIR	Near infrared response
OLI	Operational land imager

OSR	Open space ratio
UHI	Urban heat island
SEVIRI	Spinning enhanced visible and infrared imager
UAV	Unmanned aerial vehicle
RGB	Red green and blue
RMSE	Residual mean square error
RSS	Residual sum of squares
TIRS	Thermal infrared sensor
SWIR	Shortwave infrared
TOA	Top of atmospheric
L_{λ}	At-sensor spectral radiance ($W \cdot m^{-2} \cdot sr^{-1} \cdot \mu m^{-1}$)
M_L	band specific multiplicative rescaling factor
A_L	The band specific additive rescaling factor
T_B	is the brightness temperature (in degrees)
K1 and K2	coefficients determined by effective wavelength of a satellite sensor
ε	the spectral emissivity
T_s	is the LST (in Celsius)
ε_{OLI}	is the ground emissivity
λ	is the wavelength of emitted radiance
σ	is the Boltzmann's constant
h	is the Planck's constant
c	is the velocity of light

(BU), to simplify and to improve the mapping accuracy of built up areas in thematic mapper images over the conventional mapping index of built-up index (NDBI). Urban thermal field variance index (UTFVI) derived from Land Surface Temperature was adopted by (Ahmed, 2018; Singh, Kikon, & Verma, 2017) to improve characterization of urban heat island. A soil-adjusted vegetation index (SAVI) was firstly introduced by (Huete, 1988) as an efficient index in describing a dynamic soil-vegetation system from remotely sensed data. Moreover, (Chen, Zhao, Li, & Yin, 2006) used normalized difference bareness index (NDBaI) and normalized difference water index (NDWI) to enhance the mapping accuracy of barren land areas and water bodies, respectively, from remotely sensed data. (Son, Chen, Chen, Thanh, & Vuong, 2017) also developed a new index known as normalized difference composite index (NDCI), which was effective in the automatic extraction of impervious surface areas (IS) from Landsat images. Furthermore, (Zha,

Gao, & Ni, 2003), in their study on the practicality of using NDBI in automatically mapping urban areas from TM imagery, demonstrated that NDBI was a quicker and more accurate index in mapping built-up areas compared to the maximum likelihood classification method. (Adeyeri, Akinsanola, & Ishola, 2017) found that normalized difference vegetation index 705 (NDVI705) is very efficient in exploring the sensitivity of the vegetation red edge while modified soil adjusted vegetation index 2 (MSAVI2) is efficient in calculating a more reliable soil brightness correction factor in regions with uncovered soil surface. They have also defined and utilized ratio vegetation index (RVI) to improve on the existing NDVI in the area of accurate characterization of thick vegetation form sparse vegetation.

To extract essential morphological information such as vegetation area, water body, etc. for the surveyed regions, the spatial image classification technique was mainly used to extract these information



Fig. 1. Geographically referenced map of San Francisco (www.voyagesphotosmanu.com, 2019).

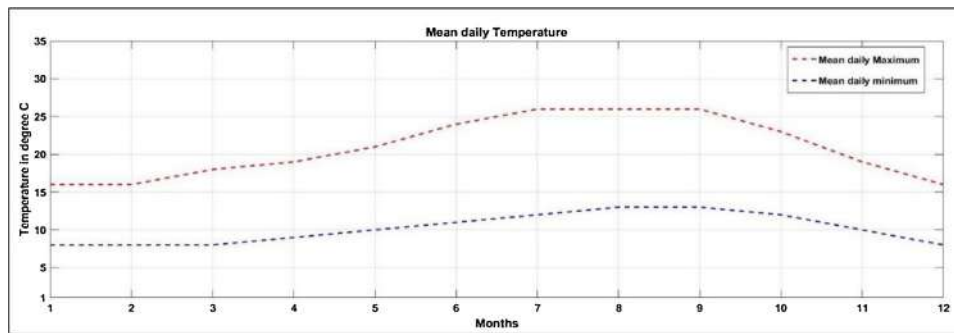


Fig. 2. Mean daily maximum and minimum temperatures for San Francisco city (Climate San Francisco International Airport, 2019).



Fig. 3. Aerial map of San Francisco and location of study areas (GoogleEarth, 2019).

Table 1

Description of area classifications.

Class	Description	Study areas
Industrial	This area comprises of industrial buildings such as warehouses, factories, etc.	A7, A12, A13, A16, A20
Commercial	This is an area with most buildings used for commercial purposes such as offices, schools, etc.	A3, A5, A8, A11, A17
Residential	This class comprises of areas with mostly residential buildings and a few basic amenities buildings such as groceries stores, etc.	A2, A4, A9, A10, A15
Recreation	This area comprises of open space areas such as recreation spaces, play grounds, parks, etc.	A1, A6, A14, 18, 19

Table 2

Description of new areas selected for model validation.

Label	Area
TA1	Commercial
TA2	Recreation
TA3	Recreation/residential
TA4	Residential
TA5	Commercial

from spatial images and to further correlate them with LST to explain the UHI intensity (Ahmed, 2018; Bokaie, Zarkesh, Arasteh, & Hosseini, 2016; Rawat & Kumar, 2015; Singh et al., 2017). The most widely used methods of classification are the supervised algorithms, including the maximum likelihood (Dewan & Yamaguchi, 2009), artificial neural networks (Pakhale & Gupta, 2010; Sangbum & Lathrop, 2006), support vector machines (Kavzoglu & Colkesen, 2009; Sun, Guo, Li, Lu, & Du, 2011) and linear un-mixing model (D. Lu, Moran, & Hetrick, 2011). The common drawback of these image classification methods is in their dependency on the consistency between the generated outcomes and trained samples. Hence, access to high-resolution spatial images are essential for achieving high accuracy results as the selected training

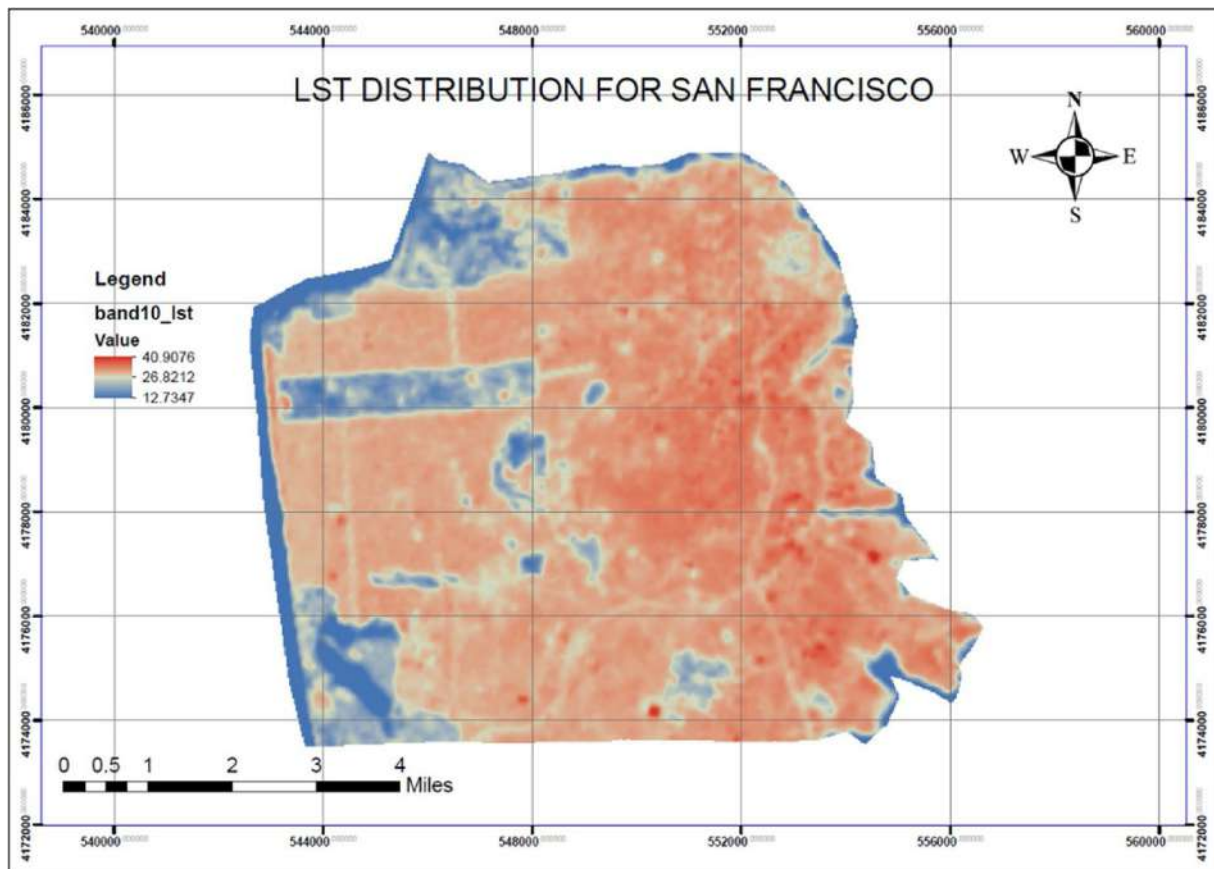


Fig. 4. Land surface temperature distribution for San Francisco city.

Table 3
Properties of spectral bands contained in Landsat 8 OLI_TIRS suit.

Landsat 8 Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS)	Bands	Wavelength (micrometres)	Resolution (meters)
	Band 1 - Ultra Blue (coastal/aerosol)	0.435–0.451	30
	Band 2 - Blue	0.452–0.512	30
	Band 3 - Green	0.533–0.590	30
	Band 4 - Red	0.636–0.673	30
	Band 5 - Near Infrared (NIR)	0.851–0.879	30
	Band 6 - Shortwave Infrared (SWIR)	1.566–1.651	30
1	Band 7 - Shortwave Infrared (SWIR)	2.107–2.294	30
2	Band 8 - Panchromatic	0.503–0.676	15
	Band 9 - Cirrus	1.363–1.384	30
	Band 10 - Thermal Infrared (TIRS) 1	10.60–11.19	100 × (30)
	Band 11 - Thermal Infrared (TIRS) 2	11.50–12.51	100 × (30)

samples need to be consistent with the classified groups.

As another limitation, satellite images obtained from LANDSAT, SEVIRI and ASTER are mainly low-resolution, and as such the resolution quality of the derived urban features (e.g. LST images and Land cover classification) are significantly influenced (Yan, Shaker, & El-Ashmawy, 2015). Techniques such as sub-pixel estimation, spatial sharpening, image fusion or regression analysis (Atkinson, 2013) have been employed in attempts to improve the image resolution. Furthermore, other innovative techniques are employed in some studies for the

improvement purpose. For example, UAV fitted with a lightweight RGB camera, a rebuild RGB camera with NIR channel, replacing the blue channel with an IR camera, was used for high-resolution spectral mapping of thermal urban properties in the municipality of Ymittos, in Athens, Greece (Gaitani et al., 2017). Nonetheless, it suffice to say that these passive aerial/satellite remote sensing techniques are still limited in generating the elevation information as this parameter is equally understood to be fundamental in explaining the UHI effect (Mirzaei & Haghighat, 2010). In other words, the morphological information derived from the horizontal spatial indices have been considered inadequate, especially with respect to the vertical component of a surveyed region. This weakness has been broadly addressed as a major drawback associated with the thermal remote sensing method used in the UHI studies (Mirzaei, 2015; Parsaee, Joybari, Mirzaei, & Haghighat, 2019).

To include the vertical spatial distribution to the account, several studies during the last decade such as (Benediktsson, Chanussot, & Fauvel, 2007; Keqi et al., 2003) proposed to benefit from the emergence of topographic airborne LiDAR data for land cover classification. Airborne LiDAR refers to laser profiling and scanning system for bathymetric and topographic applications, which gained commercially prominence in mid- 1990s (Yan et al., 2015). Unlike the 2D planimetric remote-sensing data, the explicit LiDAR data point cloud accounts for the elevation aspect of a surveyed area as it describes the 3D topographic profile of the Earth surface (Wang, 2013). Airborne LiDAR survey is widely adopted for spatial data classifications because of its accuracy in object recognition, e.g. (Germaine, 2011; Hodgson, Jensen, Tullis, Riordan, & Archer, 2003) adopted a method for urban impervious surface extraction; (Li, Jiang, Wang, & Zhang, 2013; Rottensteiner & Brieske, 2001) and (Lu, Im, Rhee, & Hodgson, 2014) also utilized LiDAR for extraction of buildings as well as classification of their type. Moreover, the use of Unmanned Aerial Vehicle (UAV) for



Fig. 5. 20 study areas of A1-A20 and the pixel segmentation.



Fig. 6. Five trail areas and the pixel segmentation.

Airborne LiDAR survey has become popular in LiDAR data capturing. For example, (Nagai, Chen, Shibasaki, Kumagai, & Ahmed, 2009) proposed an approach of UAV LiDAR system where a direct geo-referencing technique was used to process the collected LiDAR data for digital

3D modelling. As another example, (Lin, Hyyppä, & Jaakkola, 2011) used a small drone fitted with GPS/IMU, LiDAR, CCD and thermal cameras to conduct a fine-scale mapping of urban features such as tree height estimation, pole detection, road extraction and DTM generation.

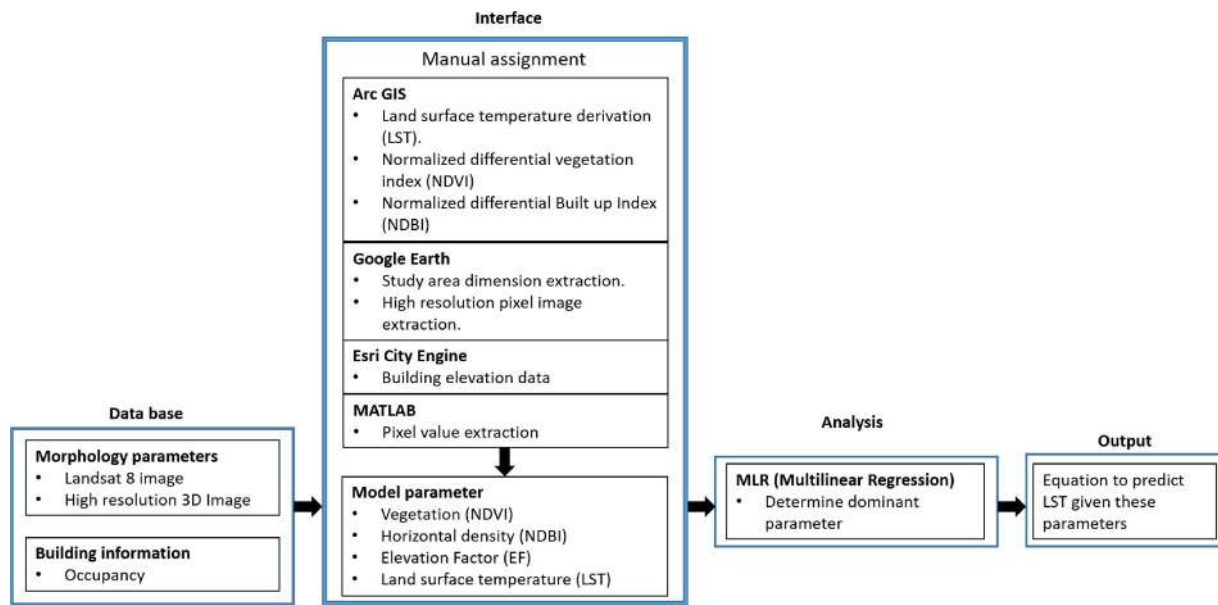


Fig. 7. Flowchart of the proposed predictive model.

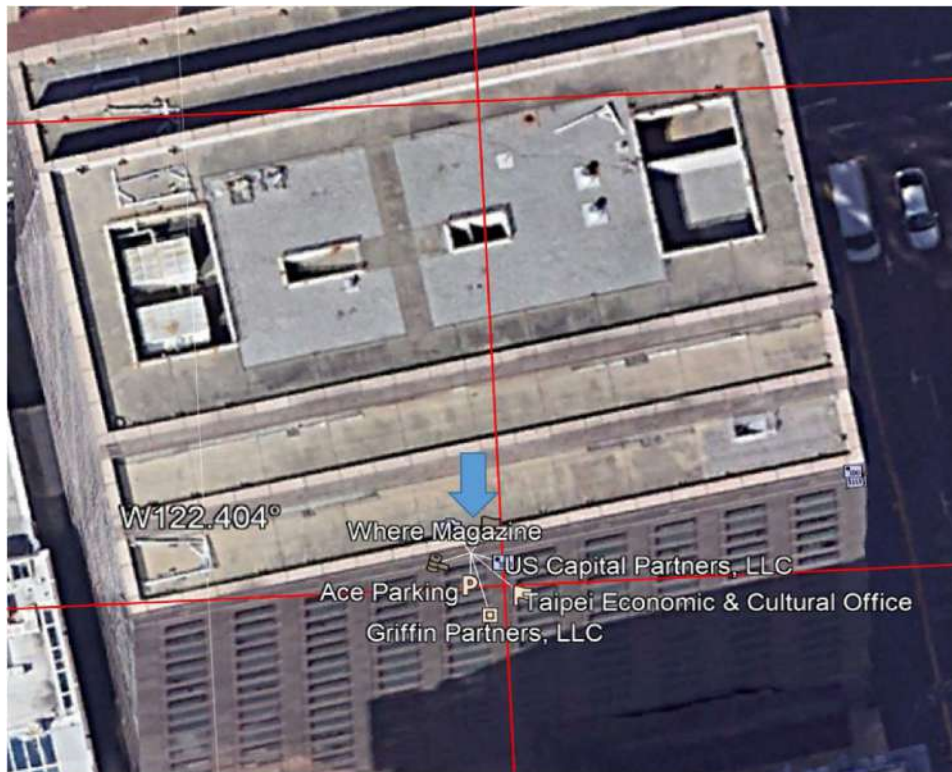


Fig. 8. Illustration of places option activated on Google earth to derive occupancy information from a building in Area 11.

Furthermore, (Lin, Hyyppä, Rosnell, Jaakkola, & Honkavaara, 2013) investigated combined use of data collected from the UAV imaging and mobile mapping systems for the land cover classification in an urban environment. Despite the mentioned advantages in using airborne LiDAR technique for generating digital terrain models (DTM), topographic mapping, and construction of digital 3D city model, the remaining issues such as expense and rigorous process involved in obtaining permits from local authorities have made this technique still an unsuitable one to provide vertical explanations of the UHI. As another alternative, in many recent studies, researchers have taken advantage of innovative and high tech 3D city modelling Geographical Information

System tools to overcome the difficulty associated with the large scale digital 3D model production.

GIS is a framework for gathering, managing and analysis geographical data to enable visualisation and to provide more information on a specific area. Based on this framework, 3D city model tools served as an interface to combine user inputted 2D spatial data set with the elevation information to generate 3D models. Such tools have become very useful in 3D city related studies especially if they are tailored for the built environment studies (Li, Quan, & Yang, 2016; Stoter et al., 2013; Zhu, Hu, Zhang, & Du, 2009), or the UHI ones, e.g. (Nakata-Osaki et al., 2018) adopted a GIS based approach to generate 3D building

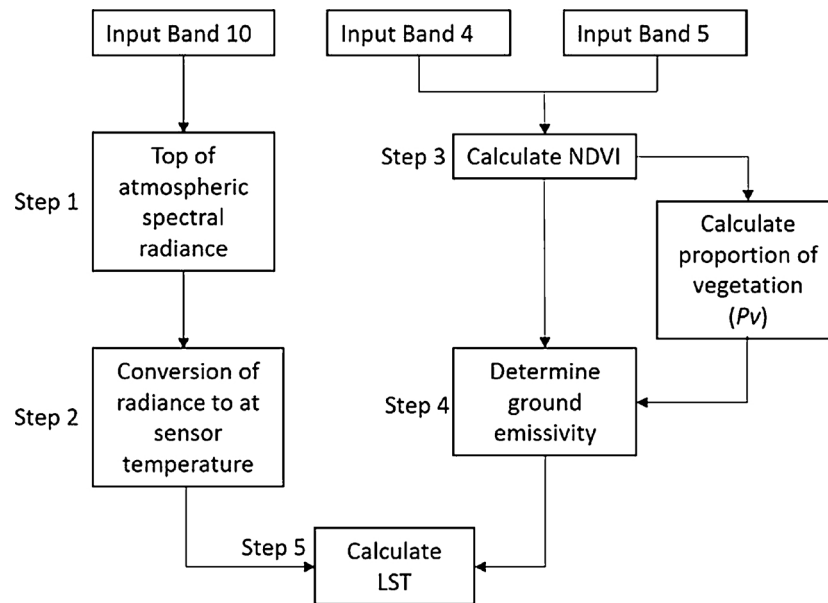


Fig. 9. Flowchart of LST retrieval from Landsat 8 OLI.

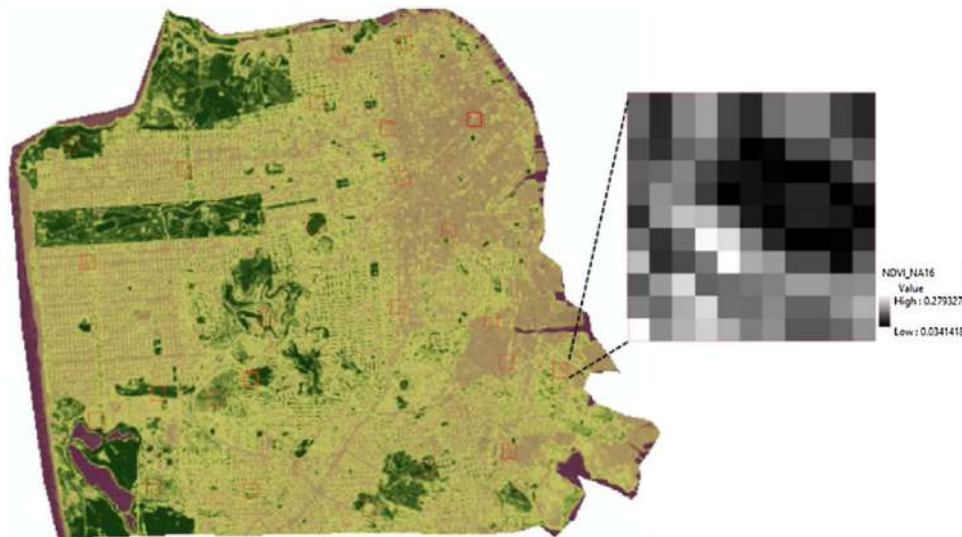


Fig. 10. NDVI distribution for San Francisco city with pixel value distribution for A16.

models for extraction of the urban geometry parameter coupled with Oke's model to determine the maximum urban heat island intensity although the vegetation impact as well as the large scale implementation of the case study were reported as the main drawbacks to this approach. Despite the described studies in the elevation extraction, there is still a need to explore approaches suitable for large scale elevation morphology extraction, to compliment the limitation of satellite based UHI approaches on the elevation information extraction, so as to improve the application of the UHI intensity explanation in large scale studies.

This study thus targets to integrate the medium-resolution Landsat image and its horizontal based morphology parameters with the elevation information available in a GIS based 3D city engine. In particular, this paper aims to explore the extent of vertical morphology components impact on the UHI intensity in urban areas by defining a novel index called the elevation factor, and furthermore to propose a UHI predictive model that account for both horizontal (Landsat derived) and vertical (elevation) morphological components of a case study city. For this purpose, LST data obtained from 20 study areas in

San-Francisco city was firstly used as the UHI indicator. Esri 3D city engine, Landsat images and Google Earth were then used to retrieve both the horizontal and vertical morphology parameters in order to explain the spatial distribution of the LST across the studied regions. A multi-linear regression analysis is employed based on these parameters to develop a predictive UHI LST models. Furthermore, the predictive models were validated using five new areas, not being initially used in the training process of them.

2. Geographical information of the study area

The city of San Francisco is selected as the study area. San Francisco is the cultural, financial and commercial hub of Northern California. According to (U.S.CensusBureau, 2017), San Francisco is ranked 13th in list of most populous city in the United States, and the 4th most populous in California with a population of 884,363 residents in 2017. It is located on the West Coast of the United States at the north end of the San Francisco Peninsula as shown in Fig. 1 and has a latitude and longitude of 37 ° 47'N; 122 ° 25'W, respectively. Its landmass is

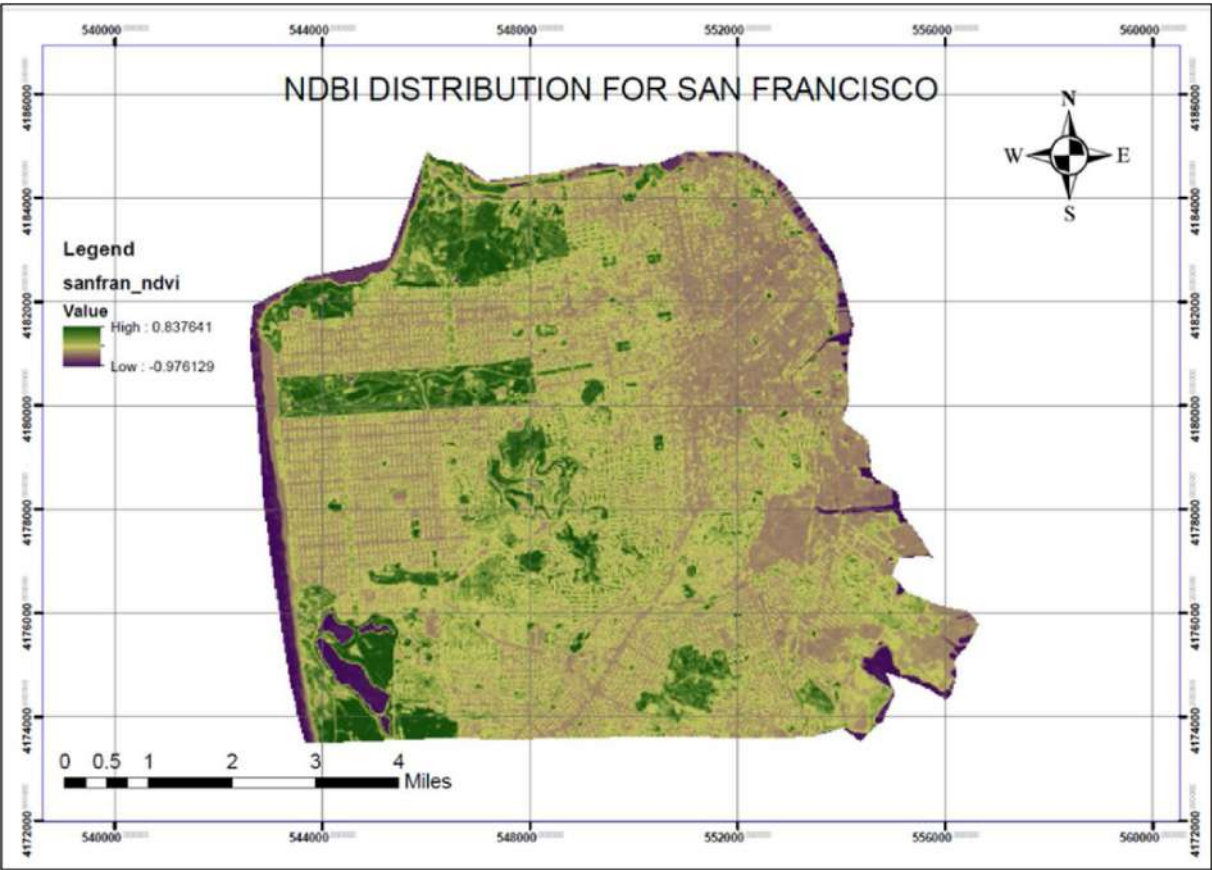


Fig. 11. NDBI distribution for San Francisco.

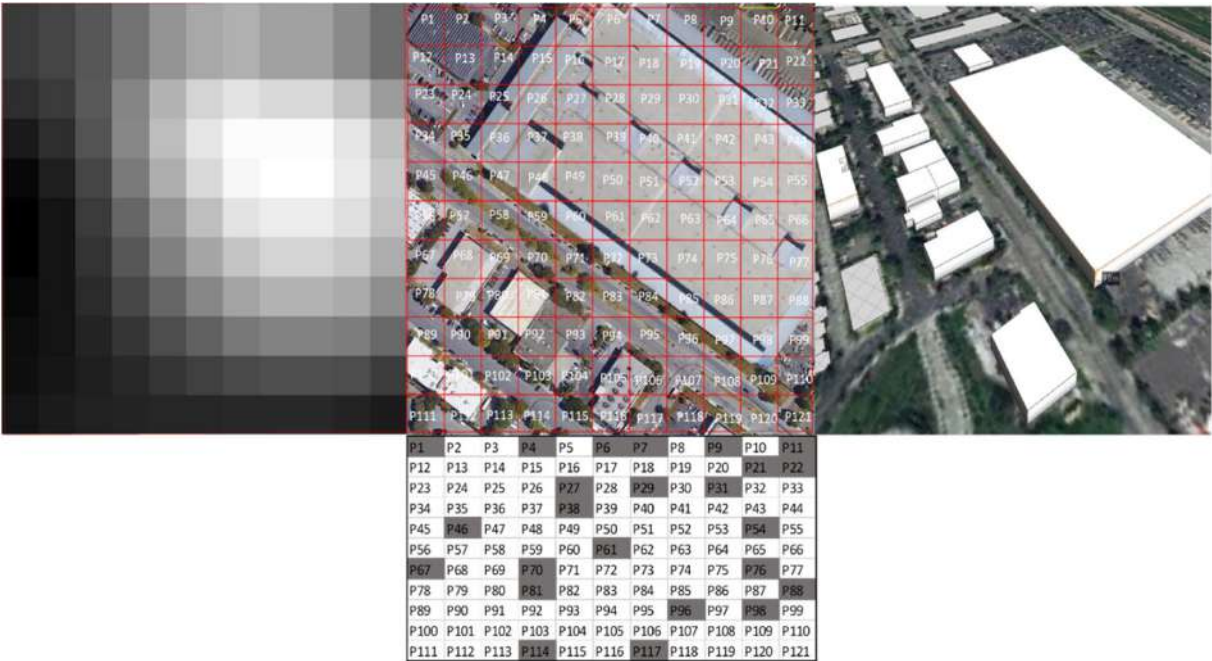


Fig. 12. Cell division with the pixel resolution of the output LST image for A16 and randomly selected pixels.

estimated at about 46.89 square miles (121.4 km²). According to the Koppen climate classification, San Francisco has a warm-summer Mediterranean climate with moist mild winters and dry summers (GoldenGateWeatherServices, 2009). San Francisco's weather is strongly influenced by the cool currents of Pacific Ocean on the west

side of the city, and the water of San Francisco Bay to the north and east, which moderates temperature swings and results in a remarkably mild year-round climate with a little seasonal temperature variation. The annual weather data obtained from the San Francisco international airport is provided in Fig. 2.

Table 4
Randomly selected pixels in each areas used for the regression analysis.

A 1	A 2	A 3	A 4	A 5	A 6	A 7	A 8	A 9	A 10	A 11	A 12	A 13	A 14	A 15	A 16	A 17	A 18	A 19	A 20
P7	P17	P3	P3	P14	P14	P4	P5	P4	P6	P4	P3	P6	P8	P2	P1	P2	P7	P9	P7
P11	P19	P5	P10	P43	P21	P10	P7	P6	P25	P25	P58	P16	P13	P7	P4	P5	P18	P11	P8
P15	P29	P8	P14	P58	P26	P21	P13	P24	P29	P68	P72	P19	P17	P11	P6	P9	P22	P17	P12
P22	P35	P11	P18	P63	P28	P26	P14	P30	P31	P98	P73	P24	P18	P16	P9	P10	P26	P19	P20
P28	P42	P14	P33	P66	P34	P29	P15	P41	P35	P112	P77	P31	P23	P21	P11	P13	P27	P20	P24
P39	P45	P18	P46	P96	P41	P31	P31	P53	P37	P20	P78	P33	P24	P23	P21	P15	P28	P34	P26
P42	P54	P23	P48	P109	P47	P34	P37	P54	P49	P31	P86	P36	P30	P42	P22	P23	P31	P36	P27
P46	P57	P27	P51	P12	P54	P38	P43	P55	P51	P39	P87	P42	P31	P43	P27	P28	P32	P38	P29
P51	P58	P29	P52	P4	P57	P44	P49	P58	P55	P56	P92	P44	P33	P46	P31	P51	P34	P42	P41
P53	P62	P30	P57	P30	P64	P48	P55	P60	P56	P11	P96	P51	P34	P57	P38	P60	P35	P46	P66
P57	P66	P33	P58	P35	P65	P57	P56	P68	P59	P12	P101	P56	P39	P63	P46	P66	P40	P51	P70
P63	P67	P35	P62	P41	P70	P64	P58	P71	P60	P13	P112	P60	P41	P71	P54	P67	P52	P55	P74
P67	P68	P37	P67	P1	P71	P73	P65	P73	P65	P30	P116	P64	P46	P73	P61	P69	P56	P56	P77
P72	P69	P45	P68	P59	P74	P74	P78	P93	P67	P34	P119	P66	P65	P74	P67	P74	P59	P73	P81
P87	P79	P53	P76	P60	P86	P75	P79	P94	P68	P66	P11	P67	P69	P78	P70	P84	P60	P93	P84
P90	P81	P57	P87	P70	P87	P84	P80	P106	P73	P70	P31	P72	P71	P79	P76	P90	P67	P97	P86
P93	P89	P61	P90	P79	P89	P99	P84	P107	P75	P71	P57	P83	P76	P91	P81	P93	P71	P102	P87
P104	P99	P71	P91	P82	P98	P101	P86	P110	P80	P73	P62	P86	P96	P100	P88	P96	P73	P106	P88
P108	P100	P89	P97	P88	P105	P103	P102	P112	P82	P84	P86	P88	P108	P101	P96	P98	P74	P111	P91
P110	P101	P107	P113	P91	P107	P104	P103	P113	P95	P94	P89	P89	P112	P112	P98	P105	P81	P115	P93
P117	P107	P110	P115	P98	P110	P30	P104	P121	P98	P106	P115	P109	P115	P5	P114	P107	P83	P117	P102
P36	P108	P114	P117	P103	P12	P17	P113	P45	P113	P113	P4	P114	P117	P80	P117	P115	P97	P121	P108
P30	P112	P121	P1	P10	P1	P91	P120	P10	P121	P116	P50	P118	P121	P120	P7	P121	P100	P3	P110
P100	P20	P77	P11	P28	P120	P121	P121	P100	P2	P13	P6	P3	P81	P28	P29	P100	P4	P10	P50



Fig. 13. Building dimension extraction interface on Esri 3D city engine for A16.

2.1. Study domain and measured parameters

San Francisco was chosen for this study because of a free access to available 3D building dimension data to be used for the geometry construction using Esri city engine satellite database. 20 study areas, denoted with the symbol of A1-A20 as shown in Fig. 3, were selected based on the area classification. For the purpose of this study, the entire city was classed into four categories (described in Table 1) based on the occupancy/use of buildings. These 20 study areas were selected in such a way that all the four outlined areas have an even number of representative (Table 2).

The LST map for San Francisco, showing the LST distribution across the region is shown in Fig. 4. In addition, five more areas were randomly selected to be later used in testing performance of the proposed model to predict the LST prediction. These areas are referred as trial

areas in this study and are denoted with the symbol TA1-TA5. Details of these five areas are given in Table 3. Each area has a dimension of 300 m × 300 m chosen to correspond with the downloaded satellite image resolution as depicted in Figs. 5 and 6.

3. Methodology

The flowchart illustrated in Fig. 7 highlights the stages involved in development of the UHI predictive model as well as the necessary input datasets leading to the LST prediction equation. The database section describes the raw data needed for the model development, including morphological parameters and building information. The interface section highlights the processes involved in the extraction of the required parameters from the raw data, which include Pixel value extraction, LST, NDBI and NDVI derivation and EF derivation. Then, the

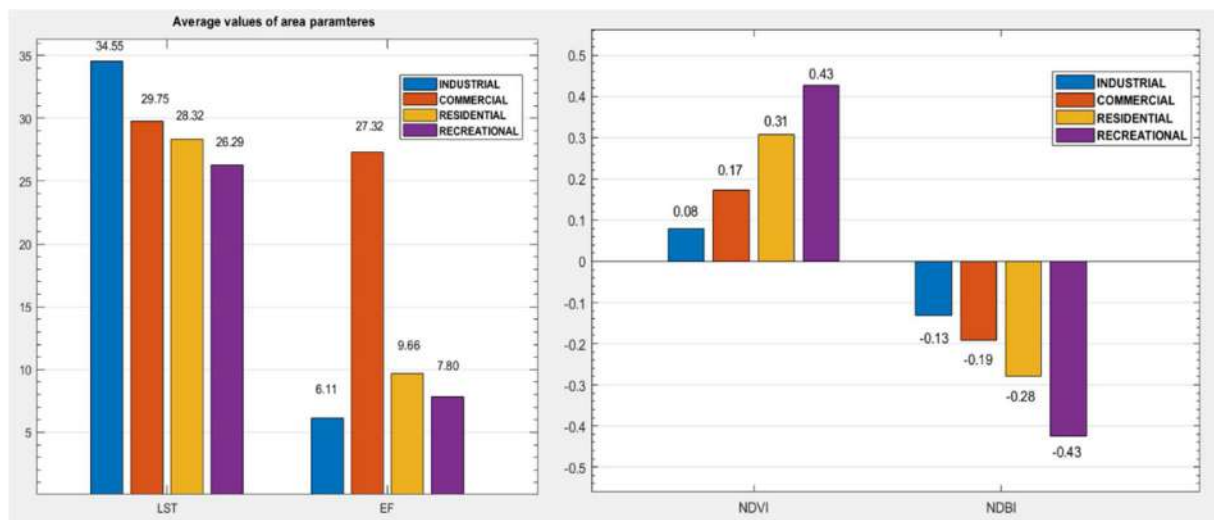


Fig. 14. Average values of considered morphological parameters for the four classes of areas.

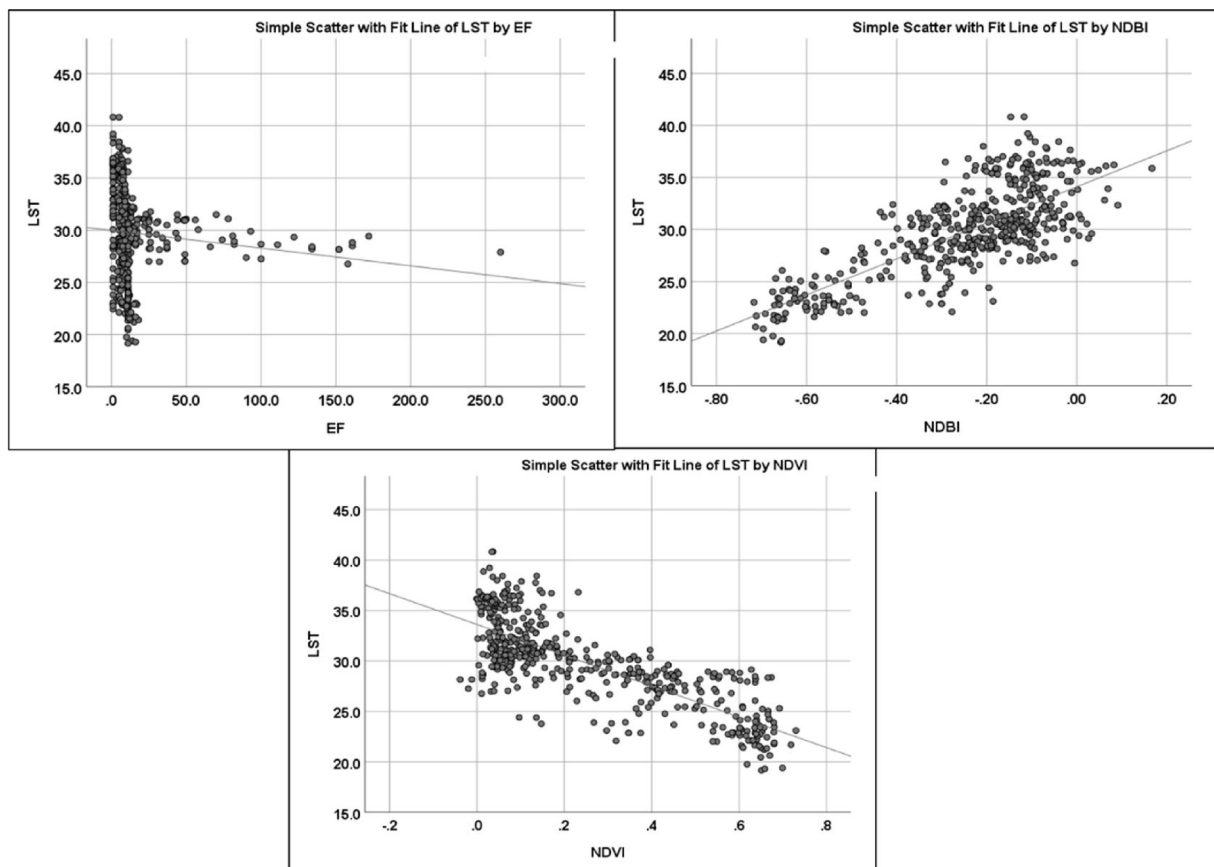


Fig. 15. Correlation scatter plots for each independent variable.

derived parameters used as the variables to train the MLR model is explained in the analysis section. Eventually, performance of the predictive equation is discussed in the result section.

3.1. Input data

3.1.1. Satellite images

As seen in Table 3, the Landsat image used for this study is from Landsat 8 OLI_TIRS suit for the city of San-Francisco, California, U.S.A acquired on 01-06-2018, at 18:44:51.9199550Z (United States

Geological Survey). This satellite data was corrected to account for the atmospheric inferences and positional errors (Roy et al., 2014; Son et al., 2017), and then was used to derive the LST, NDBI and NDVI of the selected areas. It is noteworthy to mention that TIRS bands were acquired at 100 m resolution though they were resampled to 30 m resolution in the delivered data product (Landsat.usgs.gov, 2014).

3.1.2. High-resolution 3D image

Low-resolution quality of the Landsat images has been a major drawback to its use in remote-sensing studies, especially with respect to

Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.807 ^a	.652	.650	2.537866372
a. Predictors: (Constant), NDBI, NDVI				

Fig. 16. Model information for the predictive model without EF.

Model Summary ^b					
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.826 ^a	.682	.680	2.428939637	1.018
a. Predictors: (Constant), EF, NDBI, NDVI					
b. Dependent Variable: LST					

ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	6017.672	3	2005.891	339.996	.000 ^b
	Residual	2808.280	476	5.900		
	Total	8825.952	479			
a. Dependent Variable: LST						
b. Predictors: (Constant), EF, NDBI, NDVI						

Coefficients ^a								
		Unstandardized Coefficients		Standardized Coefficients			95.0% Confidence Interval for B	
Model		B	Std. Error	Beta	t	Sig.	Lower Bound	Upper Bound
1	(Constant)	34.622	.199		173.787	.000	34.230	35.013
	NDVI	-10.822	.933	-.559	-11.600	.000	-12.655	-8.989
	NDBI	6.741	1.095	.295	6.155	.000	4.589	8.893
	EF	-.028	.004	-.175	-6.689	.000	-.037	-.020

a. Dependent Variable: LST

Fig. 17. Statistical information of the predictive model with EF (Model summary, ANOVA and Coefficient results).

the UHI as it offers limited information on the morphological features in a particular area (Yan et al., 2015). To overcome this limitation, this study obtained a high-resolution georeferenced 3D image on the study area from Google Earth and a 3D building image with all dimensions from the Esri city engine satellite database (EsriCityengine, 2019) presented with the visual image of Level of Details (LOD) 1; LOD is a 3D modelling term used to describe a 3D model that outlines the length, breadth and height of an object without giving any detail on its material or texture. These dataset, therefore, were used to extract a high-resolution morphological information for the study areas.

3.1.3. Occupancy

The occupancy information for the considered study areas was necessary to class each area according to similarity of predominant activities in that particular region. This information was obtained from Google Earth by activating the places option, which gives information on the building use at each location as demonstrated in Fig. 8.

3.2. Parameter retrieval

3.2.1. Land surface temperature (LST)

To retrieve morphological parameters like LST, NDVI, etc. from a satellite image, it has to be geo-processed. In this study, the Split-Window algorithms was adopted as the geo-processing technique to retrieve the LST of 20 studies areas. It has been proven that this algorithm is an effective procedure to retrieve the LST from Landsat 8 suits (Son et al., 2017). Flowchart shown in Fig. 9 represents the steps involved in the LST retrieval from the Landsat 8 suit and outlines the split window algorithm adopted for this purpose. More details of the flow-chart are described in the flowing sections.

3.2.1.1. Step 1 top of atmospheric (TOA) spectral radiance. Band 10, which is the Thermal Infrared band, is required to derive the thermal data like LST from the area, but first must be converted from the Digital Number (DN), which is the satellite assigned pixel value to at sensor

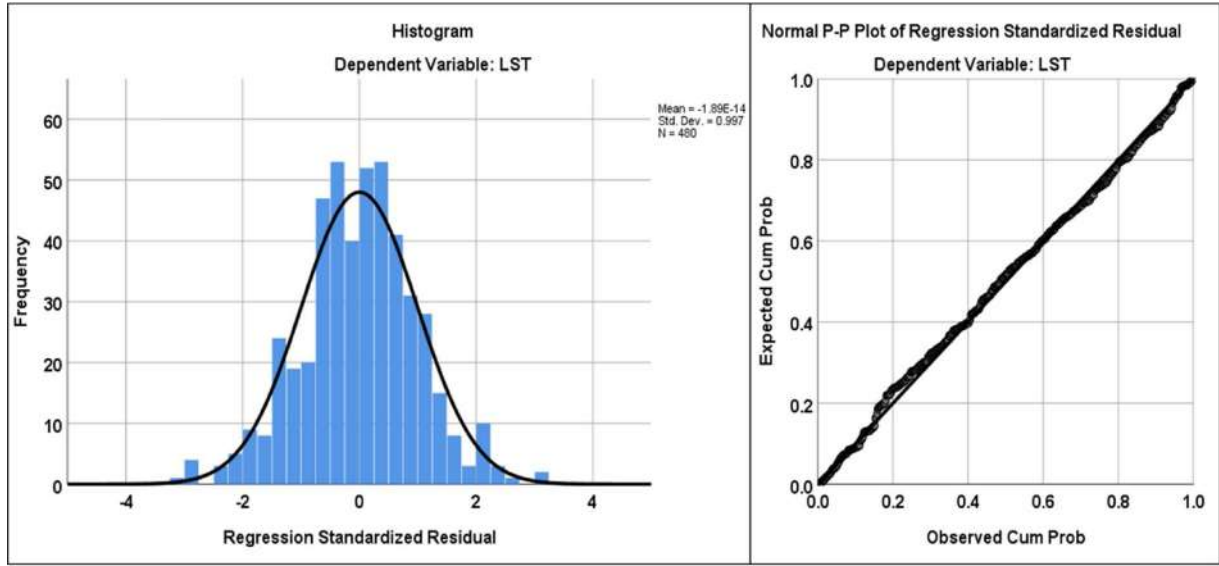


Fig. 18. Normality plot of standardized residual for the dependent variable.

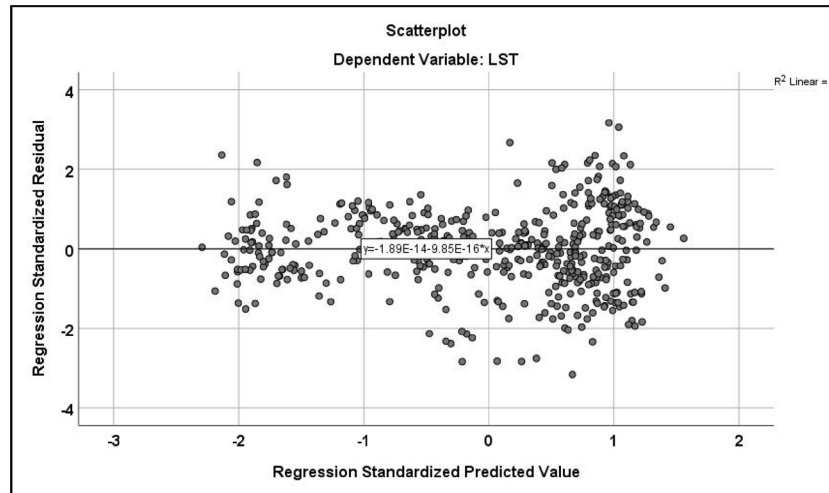


Fig. 19. Scatter plot for the standardized residual and standardised predicted value.

radiance using Eq. (1) (Kikon, Singh, Singh, & Vyas, 2016):

$$L_{\lambda} = M_L \times Q_{cal} + A_L \quad (1)$$

where L_{λ} is the at-sensor spectral radiance ($W \cdot m^{-2} \cdot sr^{-1} \cdot \mu m^{-1}$), M_L is the band specific multiplicative rescaling factor (0.0003342), Q_{cal} is the quantized calibrated values which corresponds to the Band applicable (Band 10), and A_L denotes the band specific additive rescaling factor (0.1).

3.2.1.2. Step 2 conversion of radiance to at-sensor temperature. The inverse of the Plank function is applied to the estimated radiance value to derive the temperature values (Wang, Zeng, & Karl, 1990) as below:

$$T_B = \frac{K_2}{\ln\left(\frac{K_1 \times \epsilon}{L_{\lambda}}\right) + 1} - 273.15 \quad (2)$$

where T_B is the brightness temperature (in degrees), K_1 and K_2 are the coefficients determined by effective wavelength of a satellite sensor (contained in the Meta data file), and ϵ is the spectral emissivity (typically assumed to be 0.95).

3.2.1.3. Step 3 NDVI. Parameters for LST calculation such as NDVI

index are essential in LST derivation as vegetation fraction (P_v) and ground emissivity (ϵ) are all derivatives of NDVI; hence, NDVI must be calculated and the formula is given as:

$$NDVI = \frac{NIR - R}{NIR + R} \quad (3)$$

where NIR is the Near Infrared (Band 5) and R is the Red (Band 4). In this study, ArcGIS 10.4 was used to obtain NDVI of the selected areas as depicted in Fig. 10.

3.2.1.4. Step 4 ground emissivity. The ground emissivity ϵ was calculated using the equation proposed by (Sobrino, Jiménez-Muñoz, & Paolini, 2004):

$$\epsilon = 0.00149 \times P_v + 0.986481 \quad (4)$$

where P_v is the vegetation fraction proposed by (Carlson & Ripley, 1997) and computed as follows:

$$P_v = \left(\frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min}} \right)^2 \quad (5)$$

where $NDVI_{min}$ is -0.98 and $NDVI_{max}$ is 0.83.

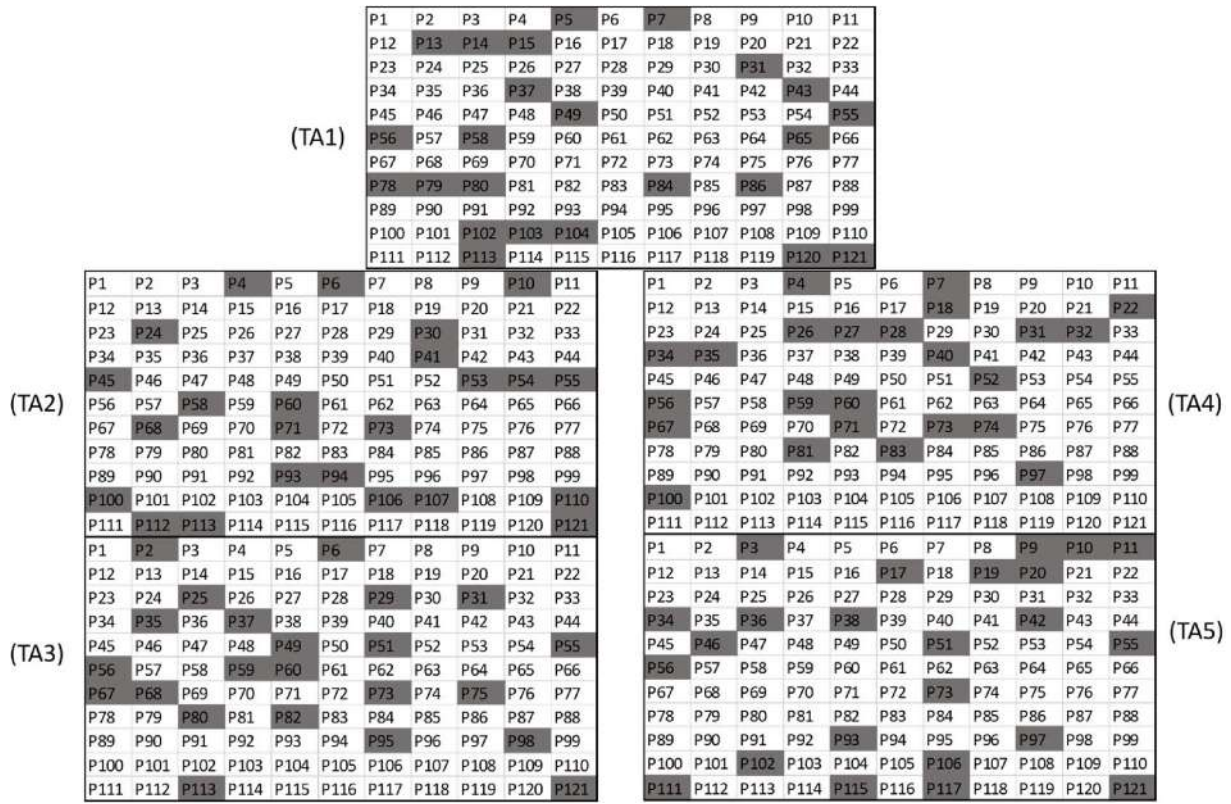


Fig. 20. 120 selected pixels from five trial areas for model validation.

Table 5
Average parameter value of five trial areas.

	EF	NDBI	NDVI	LST
TA1	7.43	−0.20	0.20	31.47
TA2	9.46	−0.31	0.34	27.74
TA3	6.53	−0.17	0.25	31.72
TA4	7.17	−0.13	0.19	29.43
TA5	4.13	−0.12	0.14	34.05

3.2.1.5. Step 5 LST calculation. The land surface temperature was retrieved from the spectral bands using the equation used by (Son et al., 2017):

$$T_s = \frac{T_B}{(1 + (\lambda \times \frac{T_B}{\rho}) \ln \epsilon_{OLI})} \quad (6)$$

where T_s is the LST (in Celsius), λ is the wavelength of emitted radiance (10.9 μ m for Landsat OLI), and ϵ_{OLI} is the ground emissivity. ρ , is the atmospheric correction factor expressed by Eq. (7).

$$\rho = h \frac{c}{\sigma} \quad (7)$$

where σ is the Boltzmann's constant (1.38×10^{-23} J/K), h is the Planck's constant (6.626×10^{-34} Js), and c is the velocity of light (2.998×10^8 m/s).

3.2.2. Normalized differential built-up index (NDBI)

The NDBI is a classification index used to map built up areas and barren ground automatically from satellite images as it has been widely used for this purpose in several studies (He et al., 2010; Zha et al., 2003). The NDBI distribution for San Francisco is presented in Fig. 11, while the formula for derivation is given as:

$$NDBI = \frac{SWIR - NIR}{SWIR + NIR} \quad (8)$$

3.2.3. Morphology parameters

Each study area was further subdivided into cells of 30×30 to correspond with the pixel sizing of the Landsat processed image files. However, due to the thickness of each dividing line, an actual grid of 11×11 cells of 27.45 m x 27.45 m dimension is obtained at the final stage. Each study area contains 121 pixels, resulting in a total pixel number of 2420 for the 20 study areas considered. In order to reduce computation time, Eq. (9) was used to determine the sample size to represent the entire areas.

$$n = \frac{Z^2 \times P(1 - P)}{e^2} / 1 + \left(\frac{Z^2 \times P(1 - P)}{e^2 N} \right) \quad (9)$$

where p is the sample proportion, which is set at 0.5. Z is the Z-score calculated, e is the margin of error, which is set at 0.04, N is the population size, which is 2420, and n is the sample size.

Based on Eq. (9), a sample size of 480 was established as representative of all 2420 pixels, hence a sampling technique was employed to randomly select 24 pixels from each of the 20 areas to make up the 480 sample size. Fig. 12 demonstrates the pixel division and the randomly selected pixels from area 16 considered for the analysis.

The morphological information of each study area is extracted from each of the randomly selected cells in the area as shown in Table 4.

3.2.3.1. LST, NDBI and NDVI pixel value extraction. The pixel values for the selected LST, NDBI and NDVI was derived using the grayscale image pixel value extraction procedure provided by (Mathworks.com, 2014). The same procedure was also adopted for the additional 5 areas used for the validation study of the predicting model.

3.2.4. Elevation Factor (EF)

Elevation Factor (EF) for each pixel is a proposed parameter to

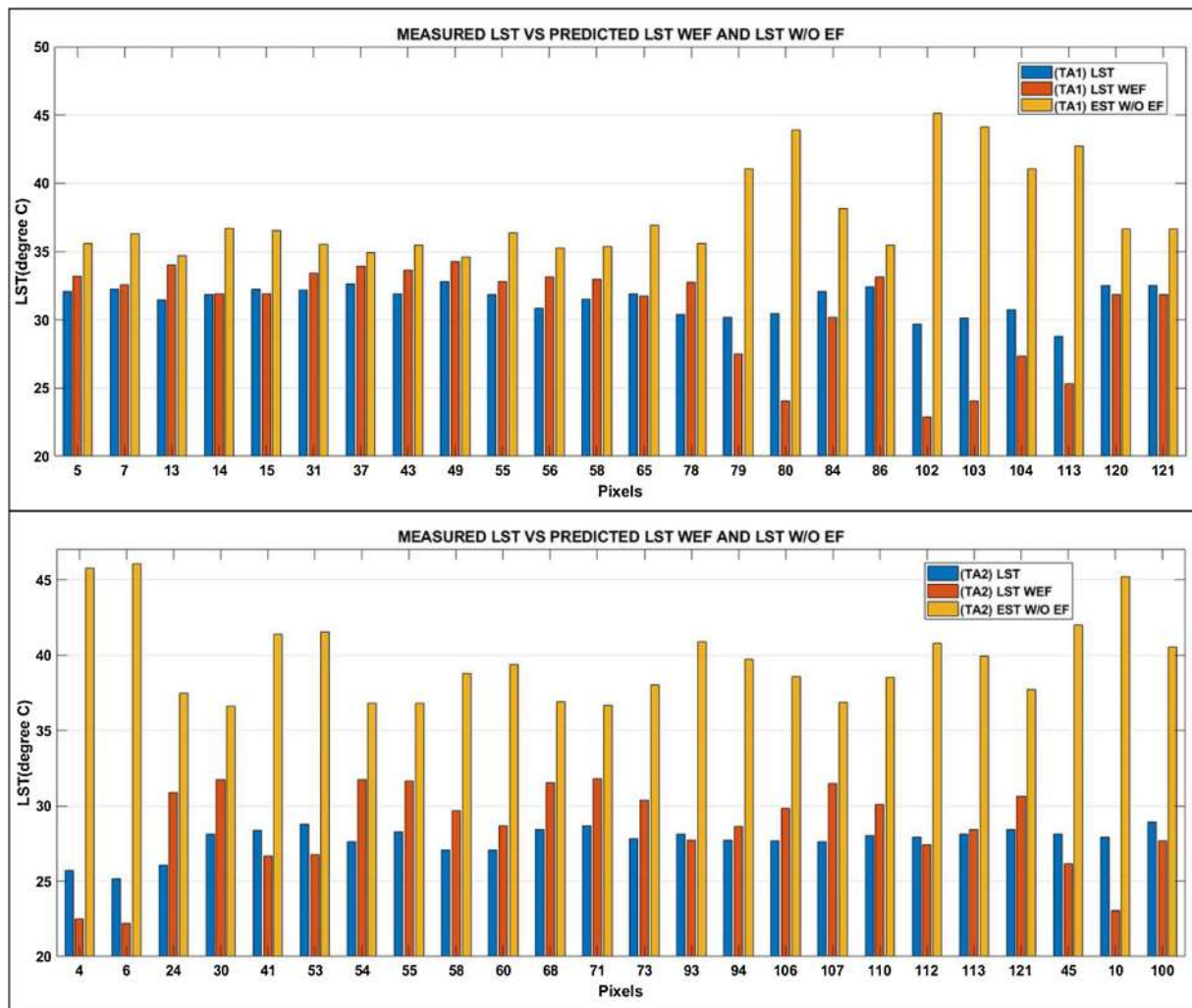


Fig. 21. performance of Models in LST prediction for Trial areas 1 and 2.

represent the overall height of the elevation feature (building and trees) within the pixel. As demonstrated in Fig. 12, each pixel/cell is shown to contain buildings, trees or both in some cases. The flat surfaces like roads or open fields are assumed to have a height value of 1 for each cell while the value from the highest elevation feature (tree/building) was adopted as the elevation factor (EF). To extract the building height, the Esri city engine software interface was used to access the building dimensions satellite dataset for the study areas as illustrated in Fig. 13.

The tree height for each area was retrieved by activating the 3D features option on Google Earth Pro software to access the high-resolution 3D virtual reality scene of the area. Using this scene, the tree heights was estimated by observation, in relation to the height of the nearest. The observed tree heights were classified as T1 (5 m height), T2 (7 m height) and T3 (11 m height) for ease of the representation.

3.2.5. Data analysis

The data was analysed using a multi-linear regression (MLR) method, which has an objective to explain the relationship between the dependent variable Y and several independent variables, assuming a linear relationship between them as expressed as below:

$$Y = \alpha + \beta_1 X_1 + \dots + \beta_k X_k \quad (10)$$

where α is the intercept, β is the slope or coefficient, and k is the number of observations.

For this study, the LST is the dependent variable (Y) while NDVI, NDBI, and Elevation Factor, constituted as the independent variables. In

more details, NDVI accounts for the vegetation density as well as the porosity of the material in an area. NDBI accounts for the horizontal density by mapping out artificial surfaces e.g. roads, buildings, barren soil, etc. It should be highlighted again that EF is a novel proposed parameter to explain the vertical density (here as building and tree heights) in an areas. The 95 % confidence level was adopted in analysing both the input and output parameters.

4. Results

4.1. Morphological parameters and study areas

Fig. 14 shows that the LST is highest at the industrial area, majorly due to the nature of the region as well as the predominant activity. The region is densely built with a large area of low height concrete buildings and tarred roads as indicated by the high NDBI value. There is also very negligible amount of vegetation also demonstrated in Fig. 14.

Commercial area is seen to be the area with the next highest LST value as seen Fig. 14. This is another densely built up area, however, unlike the industrial area, there is an increased vegetation ratio as indicated by the NDVI value due to the presence of parks and lawns. Also, the EF is in its highest value in this area as there are mostly high rise office/commercial buildings in this region, which can provide shading effect as well as a reduced sky view factor in the region caused by the tall buildings. All these factors collectively can be responsible for the lower LST value.

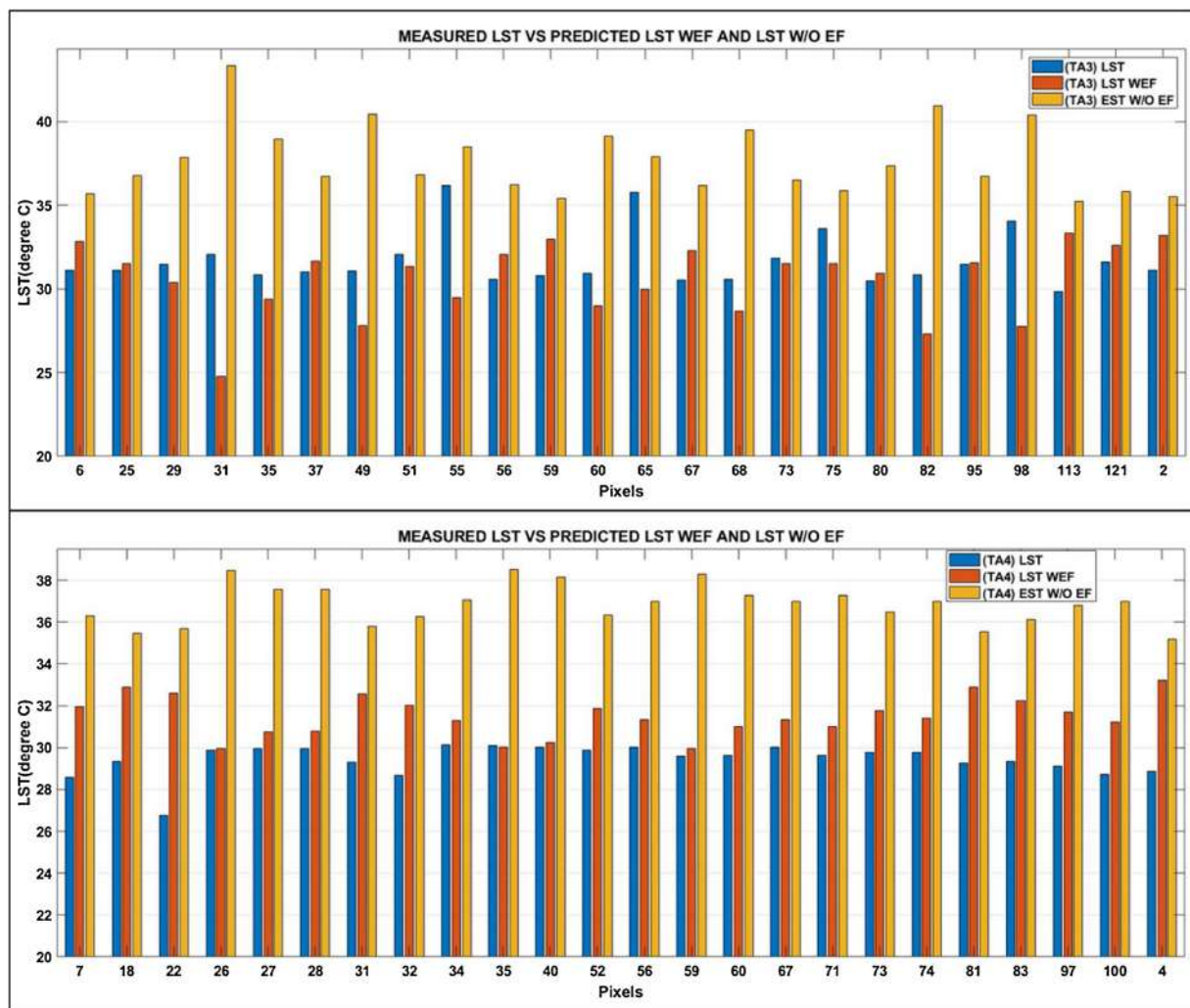


Fig. 22. Plot of measured and predicted LST for Trial areas 3 and 4.

Residential area ranks the next in LST after Industrial and commercial area. This area is fairly built up as can be seen in the NDBI value in Fig. 14. However, there is a larger vegetation area comparing to the previous two areas; there is an increased area of lawns, parks, gardens in this region. The EF is the next highest, indicating some aspect of shading effects. These factors can be responsible for its relatively low LST value.

The recreational area, is basically made up of open grass lands, thick bushes, parks sports pitches or games courts with a negligible number of low height buildings. This justifies its very high NDVI value as there is more vegetation areas with low NDBIs as there is a significantly low amount of artificial surfaces in the area. Its EF value is slightly lower than the residential area.

4.2. Regression analysis

The correlation analysis was firstly done for each of the independent variables to determine its correlation trend with the dependent variable and the results are shown in Fig. 14. Fig. 15, shows a fairly strong correlation between NDBI and the LST while an equally fairly strong negative correlation is observed between the LST and the NDVI, which agrees with findings of other studies (Ruiz, Sosa, Correa, & Cantón, 2017). However, a weak negative correlation can be observed between the elevation factor and the LST; this is possibly due to the complexities associated with capturing the elevation factors in an area as it goes beyond just buildings and trees height. Also, the fact that each pixel is

being considered as an independent entity, thereby neglecting the impact neighbouring pixels could affect its correlation. However, the EF showed a statistical significance in the regression as a predictor variable, hence, it was included in the predictive model.

4.2.1. Predictive model

The predictive model was derived from a linear regression where LST was the dependent variable and NDBI, NDVI and EF where the predictors. Firstly the regression model was conducted without adding the EF to the model and then a different regression analysis with the EF was also conducted, this was to evaluate the impact of integrating EF to predictive model.

From the model summary in Figs. 16 and 17, the coefficient of determination R^2 in the model with the EF is 0.68, which is slightly higher than the one in the model without EF which is 0.65. This implies that the model with EF can explain the variation in the data slightly better than the model without EF, as it takes the elevation aspect into consideration, as opposed to the base model, which is limited to just the elevation features.

The R value in the model with EF, which indicates the correlation between the predictors and the dependent variable, is seen to be 0.82, again emphasizing on a good correlation between parameters. The ANOVA table in Fig. 17 shows that the model is statistically significant as it has a significant value of less than 0.05. The coefficients' table also shows that all independent variables are statistically significant to the model as the p -values are 6.3×10^{-11} for EF, 1.6×10^{-9} for NDBI and 1.4

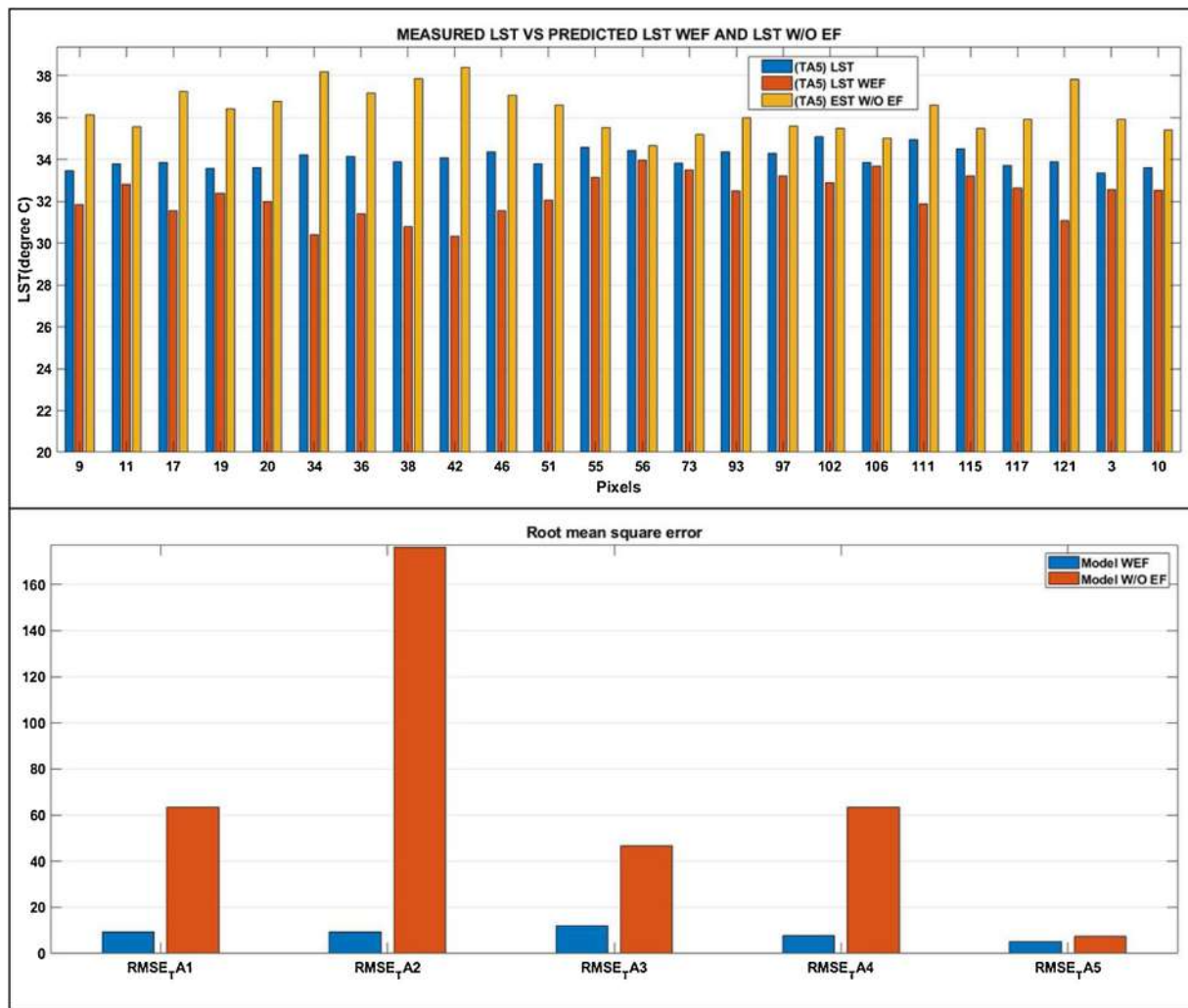


Fig. 23. Plot of measured and predicted LST for Trial area 5 and Root mean square error for the model performance in the five validation areas.

$\times 10^{-27}$ for NDVI.

Fig. 18 shows that the dependent variable (LST) is normally distributed while the normal P plot further indicates the normality of residuals as they are seen to align closely along the 45° line.

The scatter plot in Fig. 19, shows that the predictive model is statistically significant as there is no pattern in the residual and predicted standardized plot. Based on the statistical information, the prediction model equation is given as:

$$LST = 34.62 - 10.82(NDVI) + 6.74(NDBI) - 0.028(EF) \quad (11)$$

4.3. Model validation

The developed model is validated using data from 24 pixels randomly selected from the each of the five trial areas (120 pixels in total) as shown in Fig. 20.

The root mean square error (RSME) is used to measure the variance between the measured LST and the model predicted LST:

$$RMSE = \frac{\sum (y - \hat{y})^2}{n - (k + 1)} \quad (12)$$

where y and $\hat{y}$ are the measured and predicted LST respectively and n is the total number of samples. k is also the number of predictor variables. The average value of the considered morphology parameter for each selected pixel in each area is presented in Table 5.

The bar plot in Fig. 21, shows the predictive performance of the

models with EF and without EF in predicting the LST in Trial areas 1 and 2. From Fig. 21, it can be observed that the model with EF generally performed better than the model without EF in both trial areas. This implies that the elevation aspect of each pixel plays a significant role in improving the model performance in LST prediction. However, wide variations can be observed between the measured LST and the predicted ones in pixels with more complex compositions, e.g. pixels with artificial grass cover on the flat surfaces in the sport pitches, and pixels with multiple features (roads, trees and buildings) in them. This is thus evident in example pixels of 80, 102, 104 and 113 of TA1, and pixels 2, 6, 10 and 24 of TA2. Hence, it suffice to conclude that the models needs to be further improved in handling complex areas.

From Fig. 22, a similar trend of wide variations in predicted LST from measured LST, in pixels with complex composition, are equally observed in trial areas 3 and 4. Also, in these areas, the model with EF still performs better in overall than the one without EF, especially in pixels with significant elevation variations, e.g. pixels 25, 37, 73 and 95 of TA3, and 26, 27, 35, 40 and 59 of TA4.

From Fig. 23, it can be observed that although all models performed fairly better at this area as it is fairly homogenous in both horizontal and vertical features, but the model with EF performed better in overall due to its sensitivity to the elevation features in the region; hence, it is overall performance very well.

From the RMSE plot in Fig. 23, and the trial area data in Table 5, it can be seen that the model with EF performs better at areas with significant presence of elevation features as the model without EF is seen

to record the highest errors in EF dominant areas of TA1, TA2 and TA4. However, the model with EF is also seen to perform poorly in TA3 and in overall better in TA5; this is mainly due to other factors not related to EF such as complexity of the area. For instance, TA3 has an open flat surface (sport pitch) covered with artificial grass, which can impact on the LST as well. Inversely, TA5 has a fairly homogenous elevation and horizontal features, hence the low errors is recorded. In general, the discrepancies between the measured LST and the predicted models can be due to the fact that the pixels are considered as a separate entity, which does not fully represent the state of things as neighbouring pixels can have impact on the selected pixels. Also, the fact that the models only consider morphological feature in predicting LST can affect its accuracy as atmospheric and climatic factors can simultaneously have significant impact on the LST prediction.

5. Conclusion

The impact of the vertical and horizontal urban morphological components on the UHI formation and intensity was investigated in this study. Furthermore, statistically significant models capable of predicting LST in various urban areas were developed and validated. Based on the results extracted from this study, the following conclusions have been reached:

- Vertical components have significant impact on the UHI formation and intensity.
- The new proposed elevation factor (EF) shows a statistically significant *P* value in the regression analysis.
- The model performs better in areas of high elevation density in comparison to areas with low elevation density.
- The relationship between EF and the dependent variable is not very linear.
- The EF and NDVI have a negative correlation with LST while NDBI demonstrates a positive correlation with LST.

Further works to improve this study would include be the consideration of other elevation parameters besides the EF as predictor variables, as well as the development of an Artificial Neural Network (ANN) predictive model as this is been widely associated with high prediction performance. Also the atmospheric and climatic factor not considered in the development of this model should also be integrated as this would definitely have an impact on the LST.

Declaration of Competing Interest

None.

Acknowledgment

The authors would like to express their gratitude to the faculty of the engineering of the University of Nottingham for their support.

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